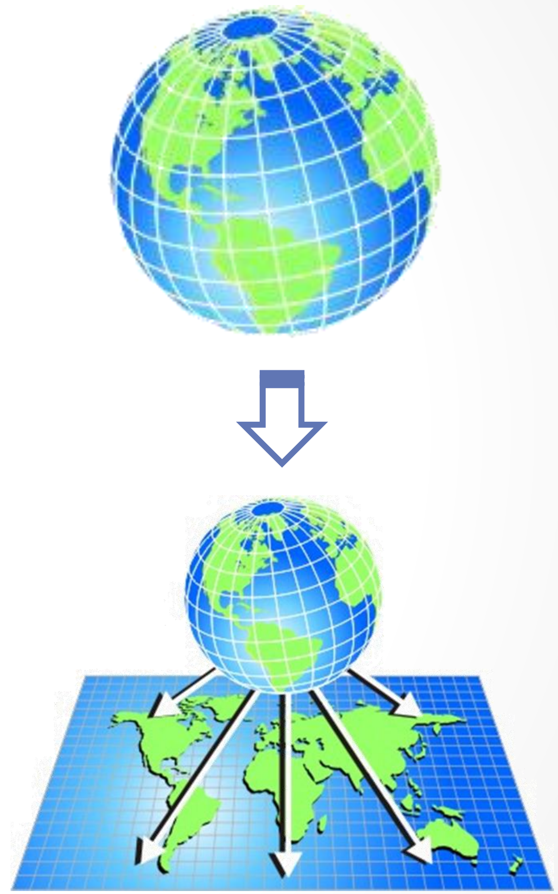
MAP PROJECTIONS

Map Projection

- Method to represent three dimensional surface of the earth on a flat 2-D surface

- Map projection is a systematic transformation of the latitudes and longitudes of locations on the surface of a sphere or an ellipsoid into locations on a plane (*An album of map projections*, 1989)

- Project from the Earth's surface to a developable surface such as a cylinder or cone, and then to unroll the surface into a plane

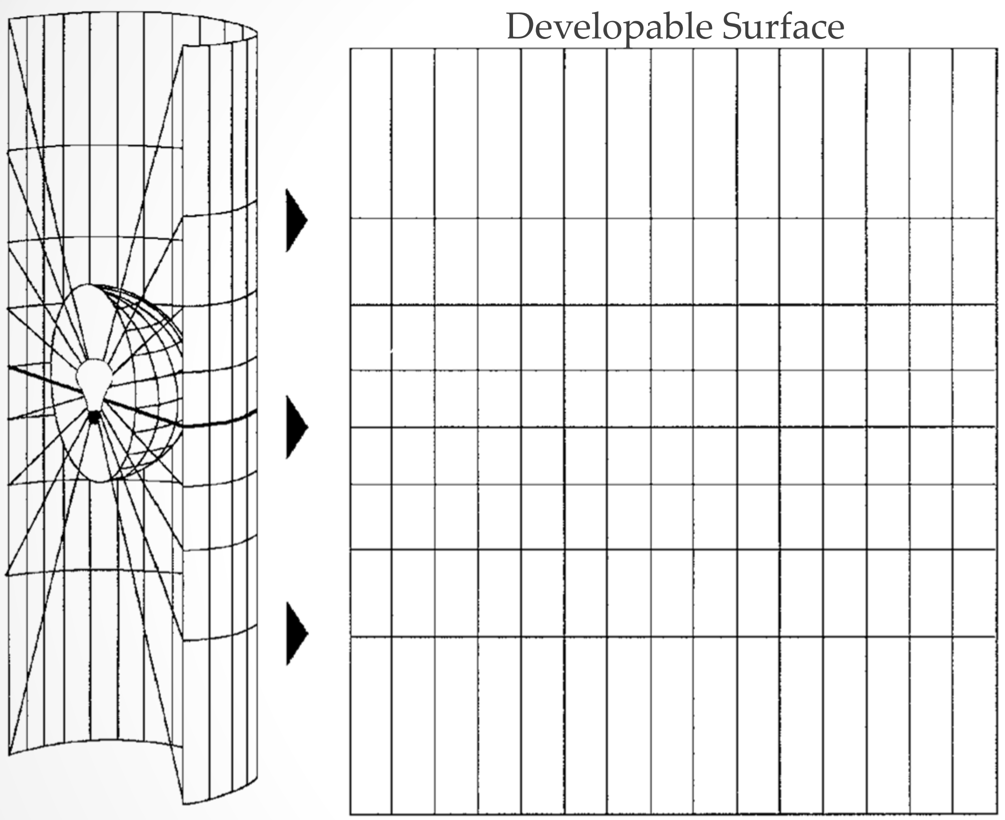


http://www.statcan.gc.ca/pub/92-195-x/2011001/other-autre/mapproj-projcarte/m-c-eng.htm

Map projections

- **Geographic Co-ordinate System**
- **Datum**
- Hence, we are now able to uniquely identify any position on Earth's surface by a pair of <u>Latitude and Longitude</u>

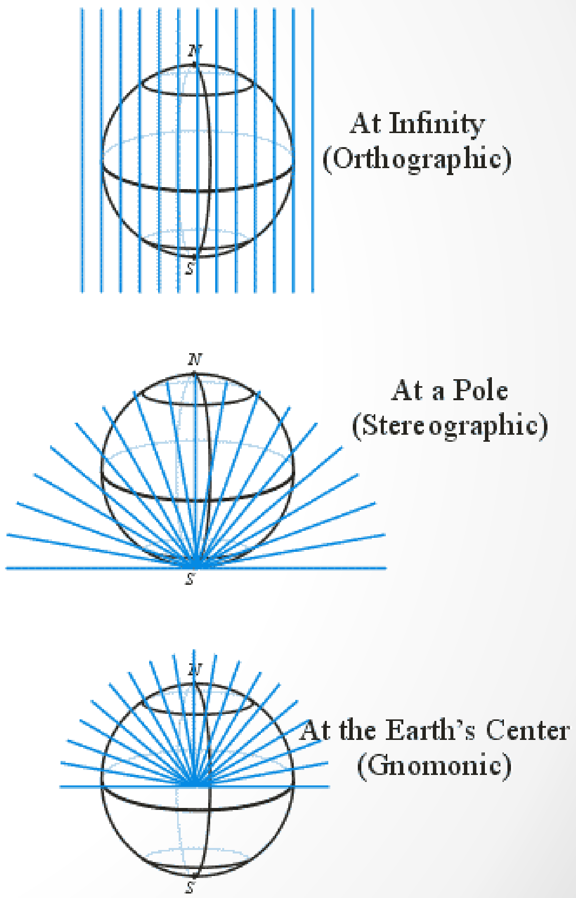
Map Projection



The graticule of a geographic coordinate system is projected onto a cylindrical projection surface.

Geographical Coordinate System (ϕ, λ) $\xrightarrow{\text{Map Projection}}$ Projected Coordinate System (x, y)

Location of Light Source



Map Projection

Mercator Projection: Conformal Preserve (angle)
Mollweide Projection: Area Preserve
Winkel Tripel Projection: Compromise

same

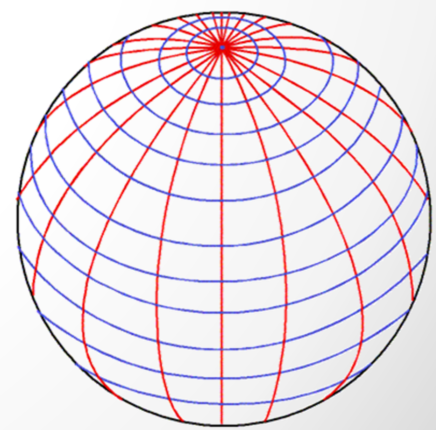
different

http://en.wikipedia.org/wiki/Mercator_projection

Light Source Positions:

1. Gnomonic at center
2. Stereographic at pole
3. Orthographic at infinity

Longitude meet at poles
and
Latitudes are ≈ equidistant



Distortion

- Earth's surface cannot be represented on a plane 2-D map without distortion

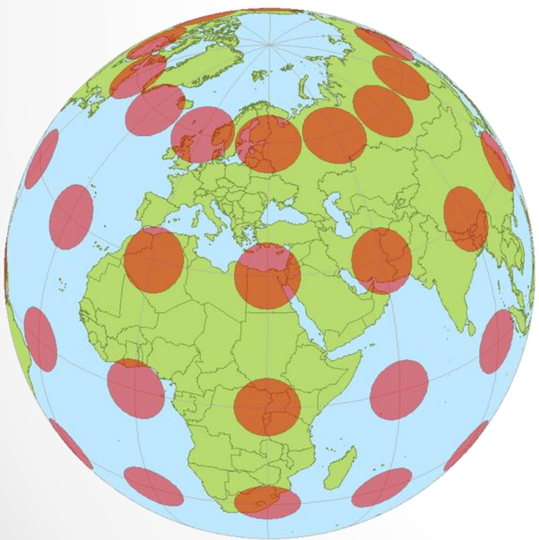
- Distortion occurs w.r.t.
  - Area – *Area are shown correctly*
  - Shape – *local shapes are shown correctly*
  - Distance – *preserves distance along some points or lines*
  - Direction – *all directions are shown correctly relative to the centre*

- A map can never preserve all of these properties at once

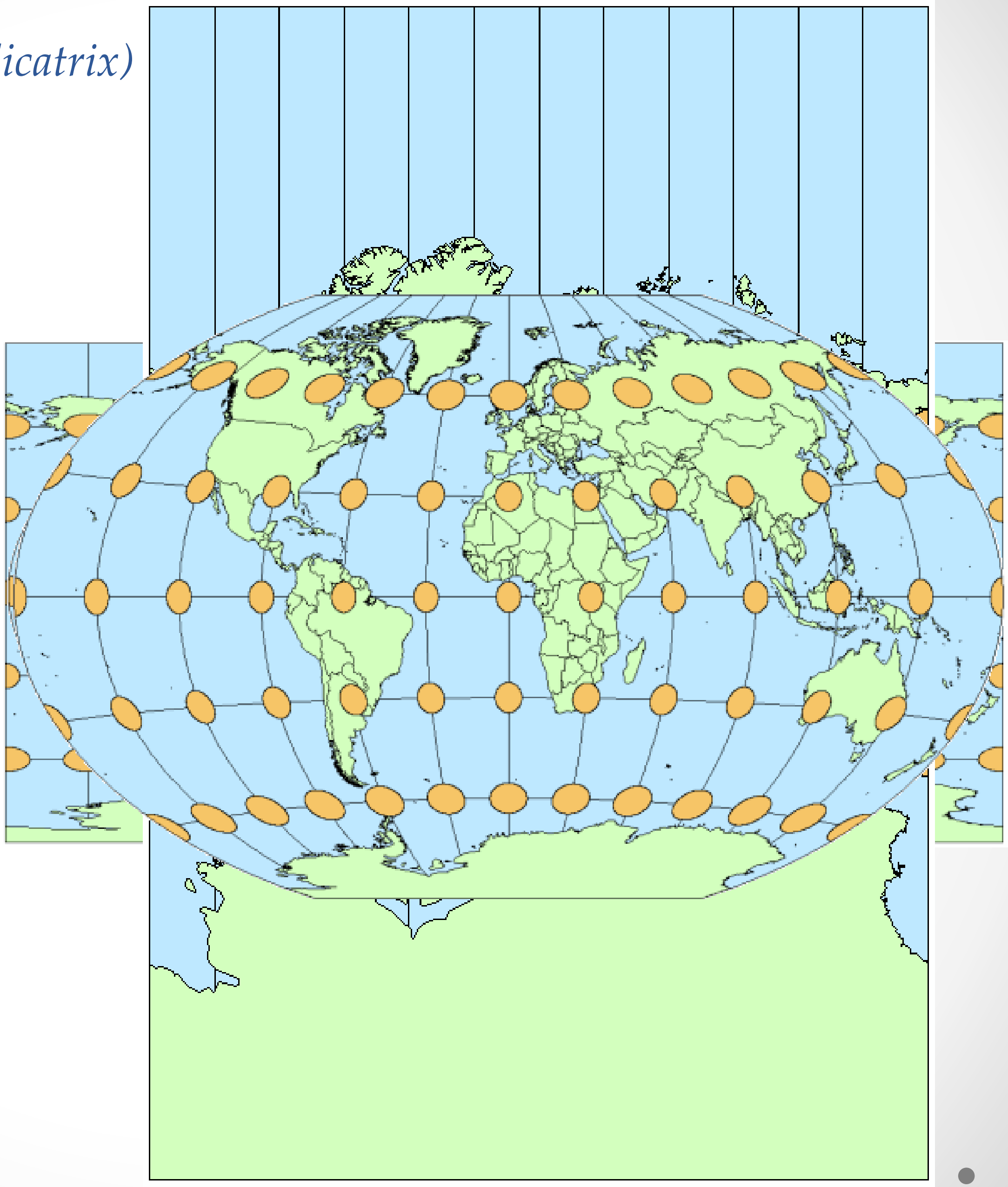
- Also, a compromise can be made among such that none of these properties are truly preserved

Distortion *(Tissot's indicatrix)*

Tissot's Indicatrix is graphical tool which helps us visualize the distortions



Compromise
Shape preserved
Area preserved

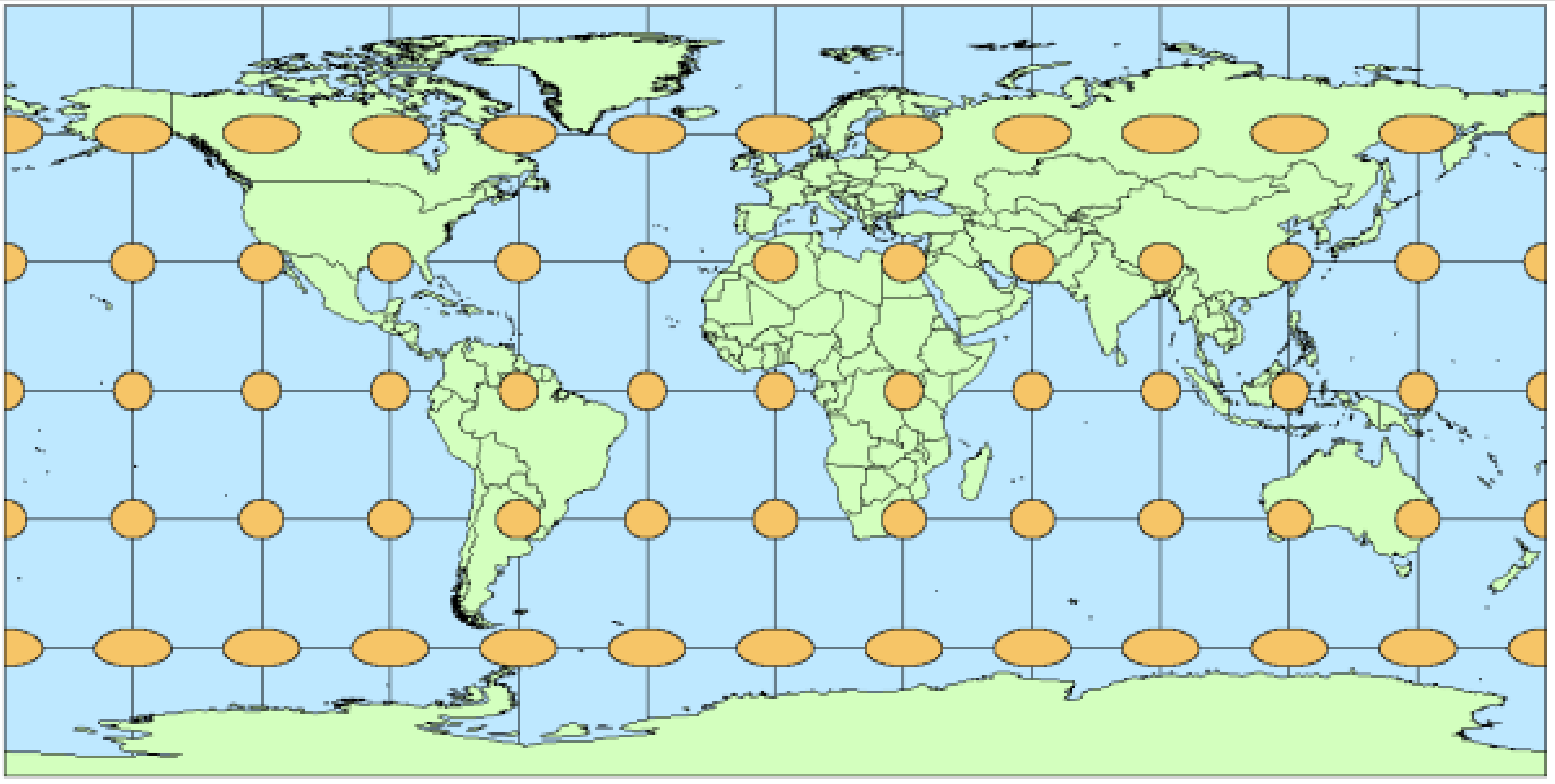


Tissot’s indicatrix was first developed by French mathematician Nicolas Auguste Tissot in 1859 as a tool to characterize the distortion due to projecting a spherical representation of the earth onto a flat surface, i.e., a map projection.

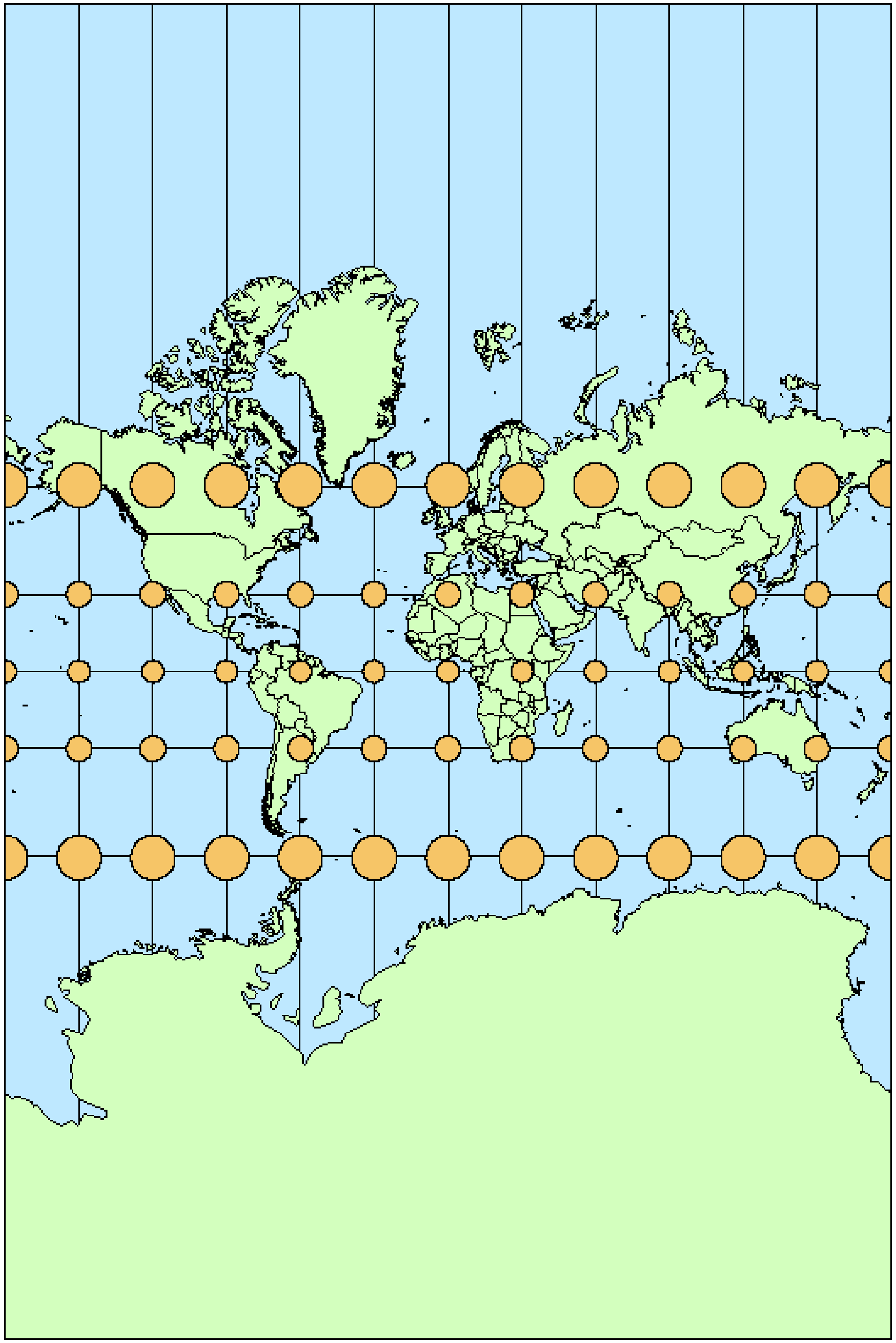
A single ellipse is called an “indicatrix”, and it shows the distortion at the point where it is centered. Because scale distortion varies across a map, usually Tissot’s indicatrices (the plural of indicatrix) are repeated across a map at regular intervals to illustrate the spatial pattern of distortion.

It is common to place them at the intersections of parallels and meridians. In the above example, indicatrices are placed at the intersections of the 30 degree parallels and meridians that make up the graticule between 60 degrees north and south.

Tissot’s indicatrix is a valuable tool in understanding and teaching about map projections, both to illustrate linear, angular, and areal distortion and to show graphically the calculations of the magnitude of distortion at each point.

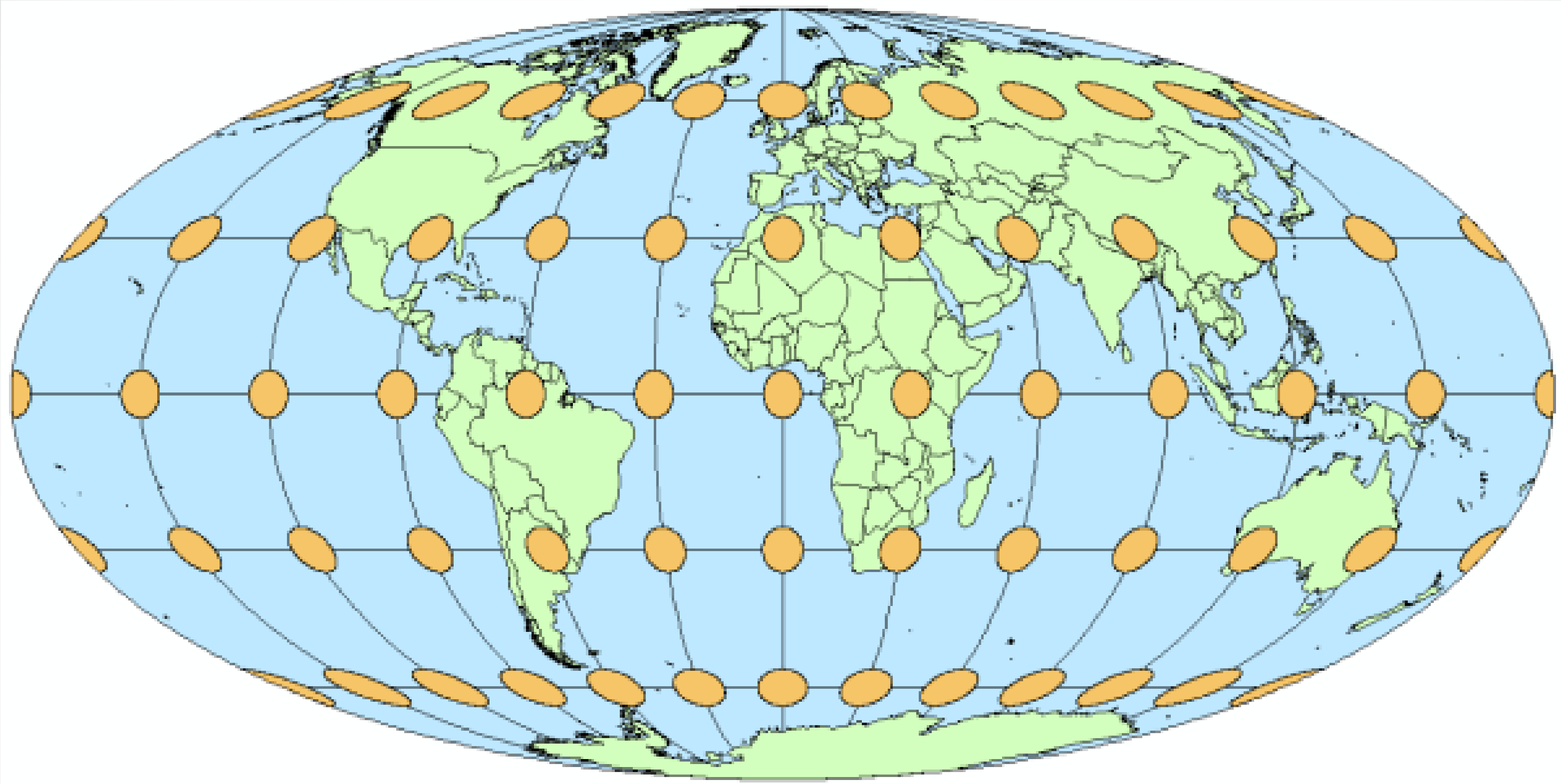


Tissot's indicatrices are used to show the pattern of distortion due to the use of the WGS84 geographic (Plate Carree) projection.



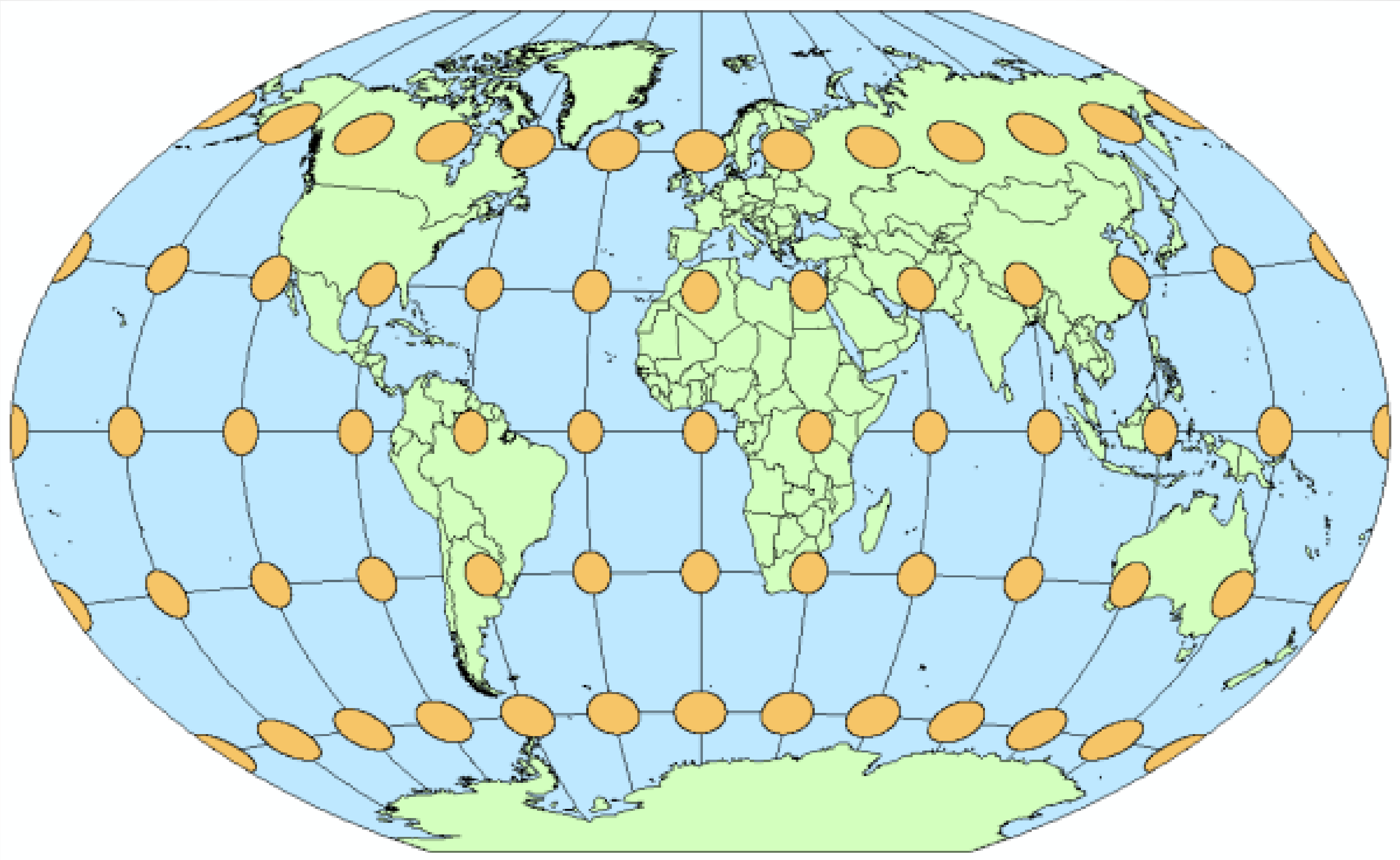
On this map with the Mercator projection, all indicatrices are circles because a conformal projection is used.

On maps that use a conformal projection, in which each point preserves angles projected from the geometric model, the Tissot's indicatrices are all circles. Because it is not possible to preserve angles and shapes at the same time, the apparent sizes of the ellipses will vary across the map.



On this map with the Mollweide projection, all indicatrices are apparently equal in size because an equal area projection is used.

On maps that use an equal area projection, in which the property of equivalence is preserved, the relative sizes of the indicatrices are retained. But equal area projections cannot also be conformal, so the ellipses will be distorted in shape and orientation across the map.



On this map with the Winkel Tripel projection, the indicatrices vary in size, shape, and orientation because a compromise projection is used.

For other map projections, such as *compromise projections*, which are neither equal area nor conformal, the area, shape, and orientation of the indicatrices may vary across the map

Different Classifications

Developable Surface

- Azimuthal
- Conic
- Cylindrical

Light Source Location

- Orthographic
- Stereographic
- Gnomonic

Properties Preserved

- Conformal
(angle or bearing preserved)
- Homolographic
(Equal-Area or area preserved)
- Orthomorphic
(shape-preserved)
- Isometric
(scaled distance-preserved)
- Compromise

Orientation of Developable Surface

- Normal
- Transverse
- Oblique

$\underbrace{\hphantom{Transverse}}$ Cylindrical / Conic

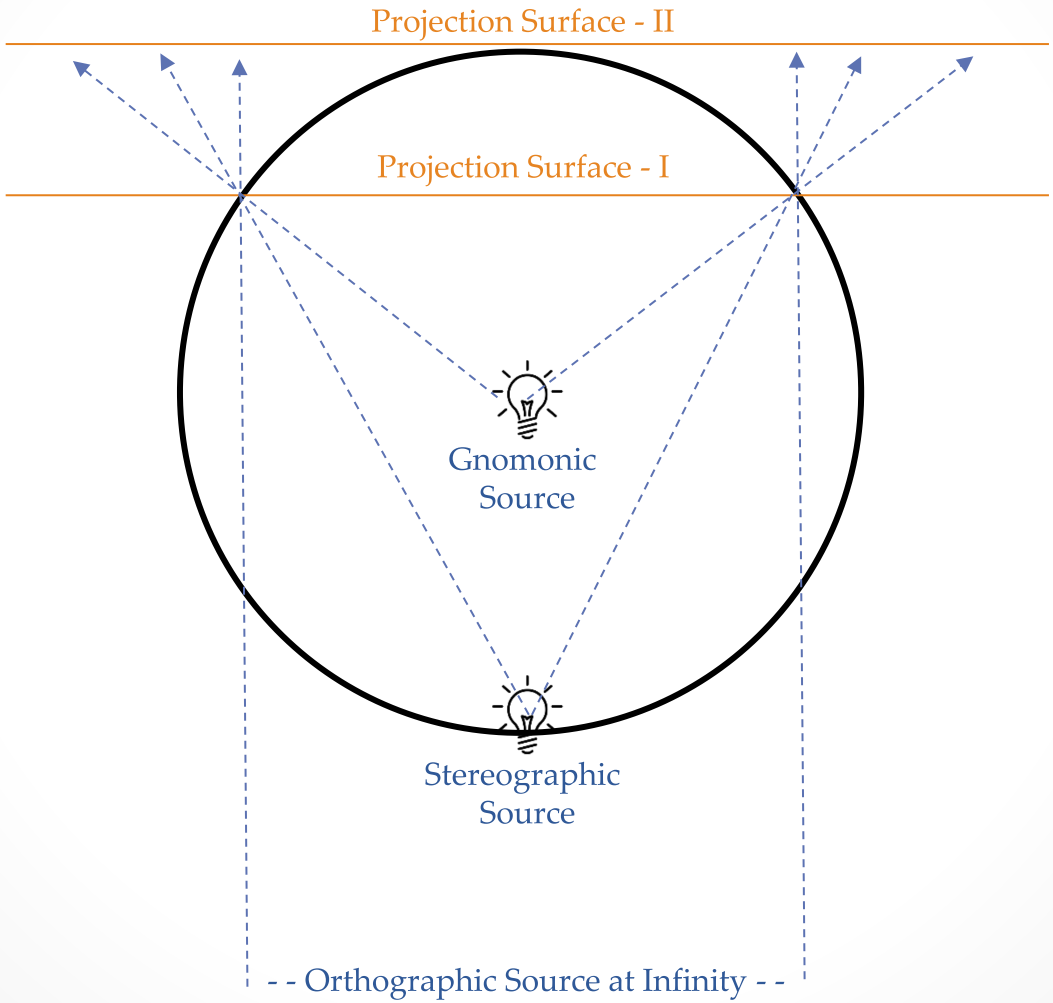
- Equatorial
- Polar
- Oblique

$\underbrace{\hphantom{Equatorial}}$ Azimuthal

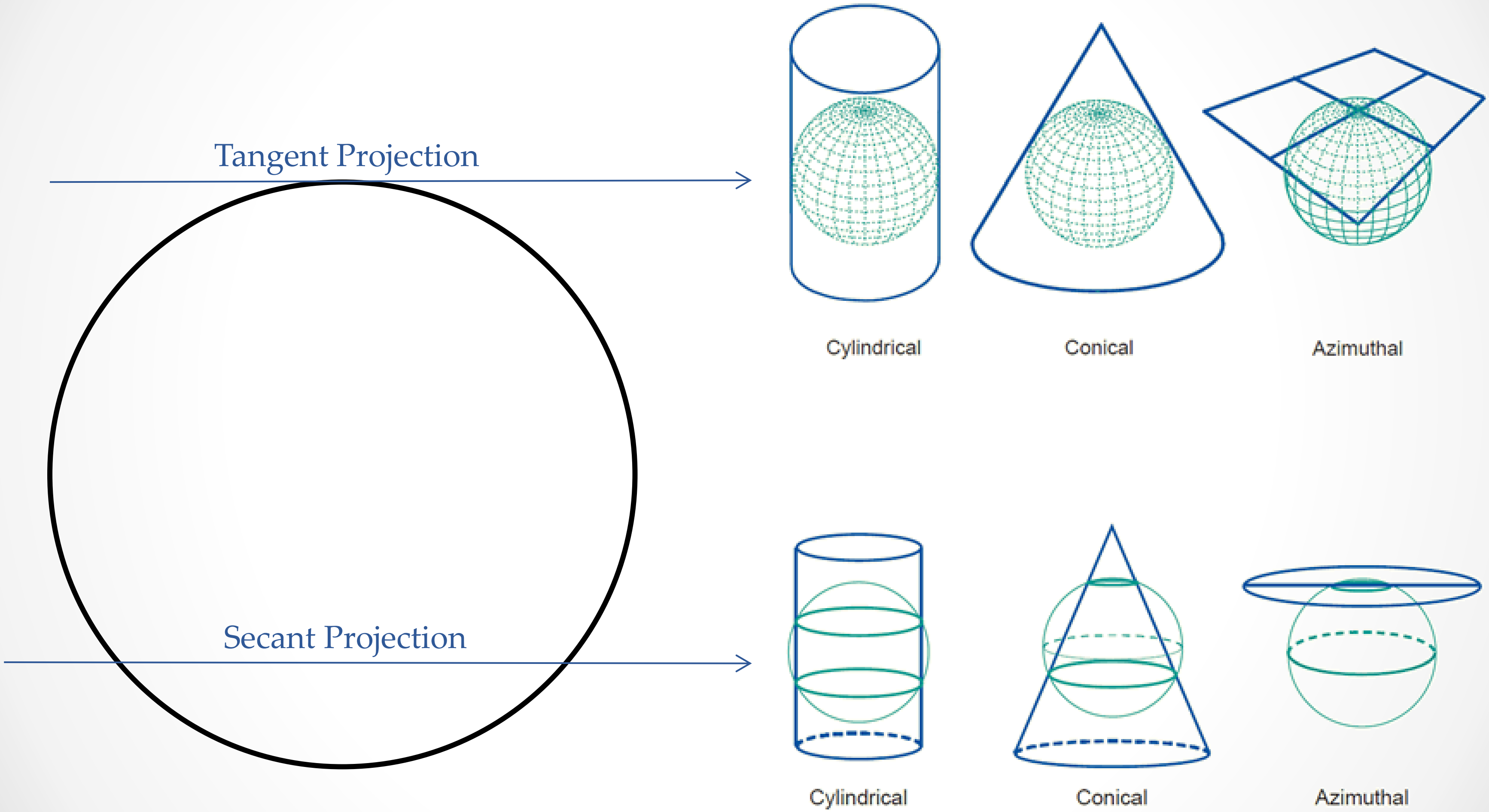
Contact with Developable Surface

- Tangent
- Secant

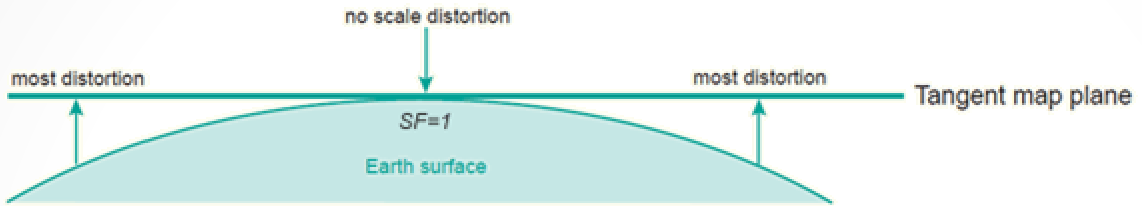
Classification – Light Source



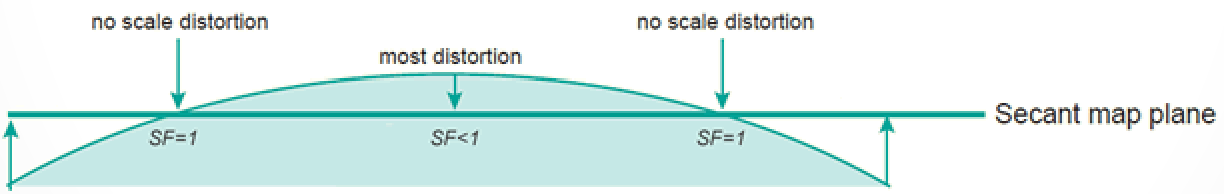
Classification – Contact



Classification – Contact



- The line of tangency between a projection surface and the surface of the globe is called the **standard line** of the projection. Along the standard line there is no distortion.

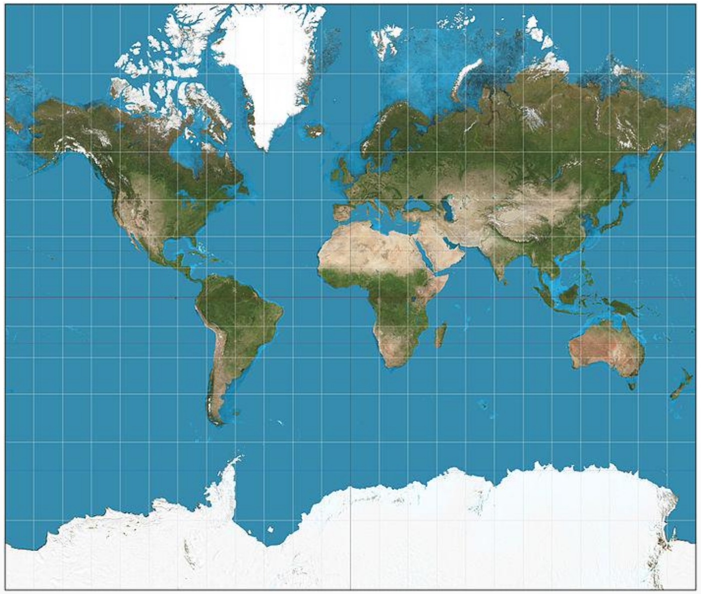


- When the projection surface intersects the globe you get a secant projection (with 2 standard lines)

Classification – Properties Preserved

1. Conformal (orthomorphic)

- Preserves local shape by using correct angles
- The angle between them is the same as between the intersection of any two lines on the map and their counterparts on the sphere; latitude & longitude are perpendicular to each other
- Eg: *Mercator*, *Stereographic, Lambert conformal conic*

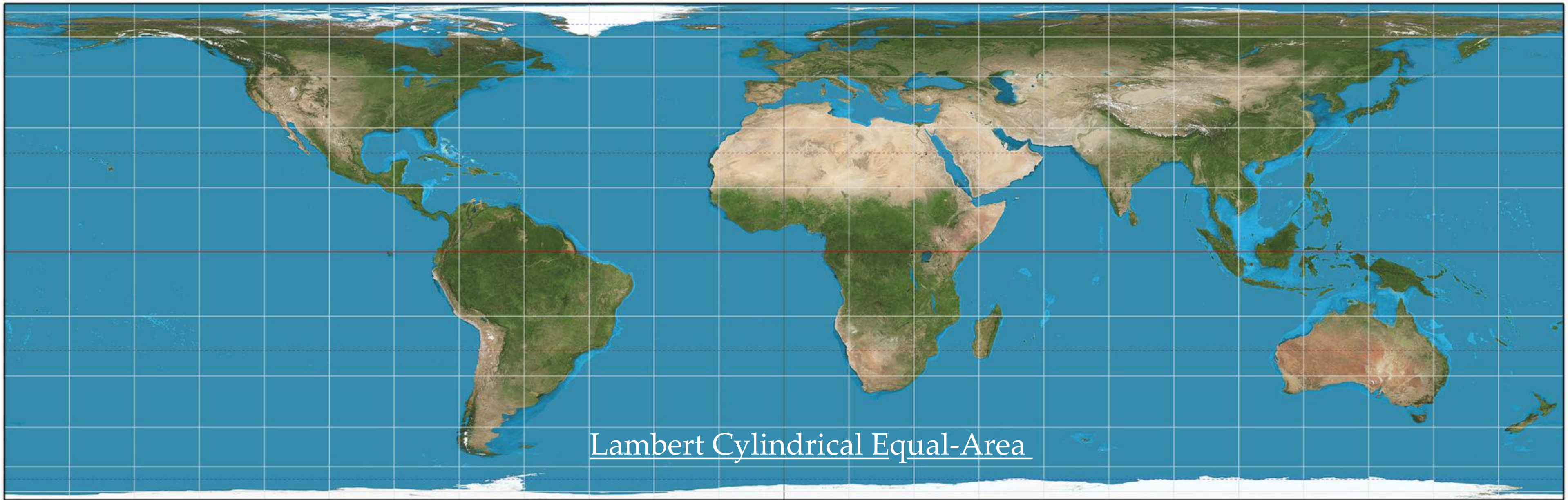


Mercator

Classification – Properties Preserved

2. Equal-Area (homolographic)

- Ratio of areas on the earth and on the map are constant
- Commonly used in GIS because of importance of area measurements
- Used for thematic or distribution maps
- Eg: *Alber's conic, Lambert's azimuthal,* *Lambert cylindrical equal-area*

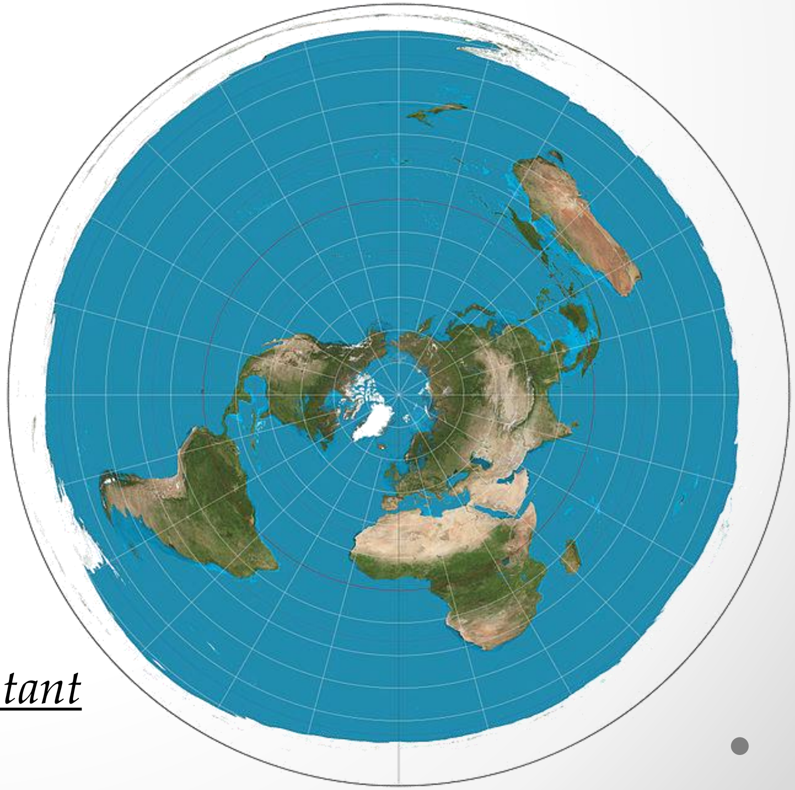


Lambert Cylindrical Equal-Area

Classification – Properties Preserved

3. Equidistant

- Preserves distance (scale) between some points or along some line(s)
- No map is equidistant (i.e. has correct scale) everywhere on map
- Important for long distance navigation
- Eg: *Azimuthal Equidistant*

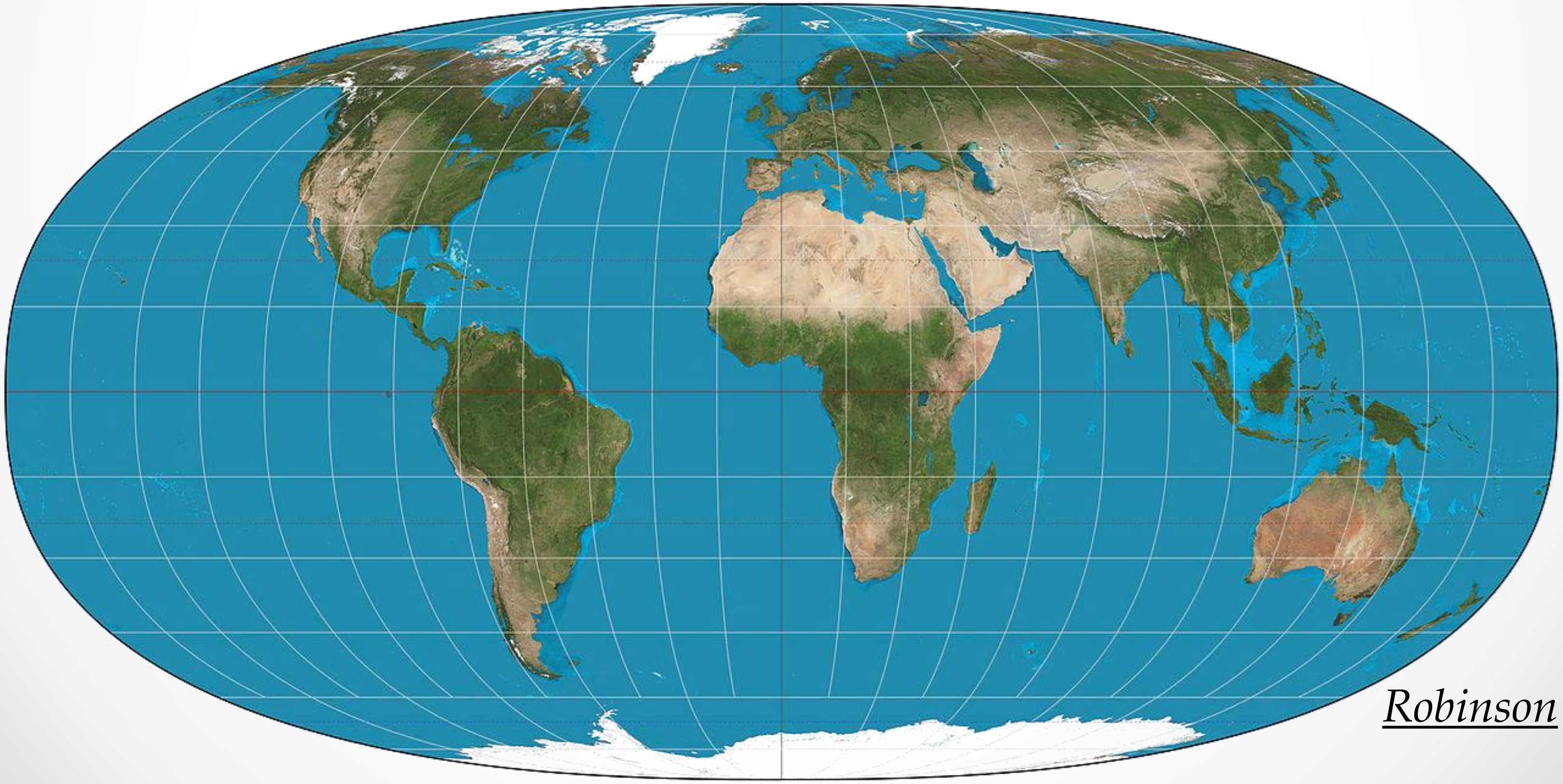


Azimuthal Equidistant

Classification – Properties Preserved

4. Compromise

- This projection system tries to seek a balance between distortions
- Eg: _Robinson_, *Miller cylindrical*

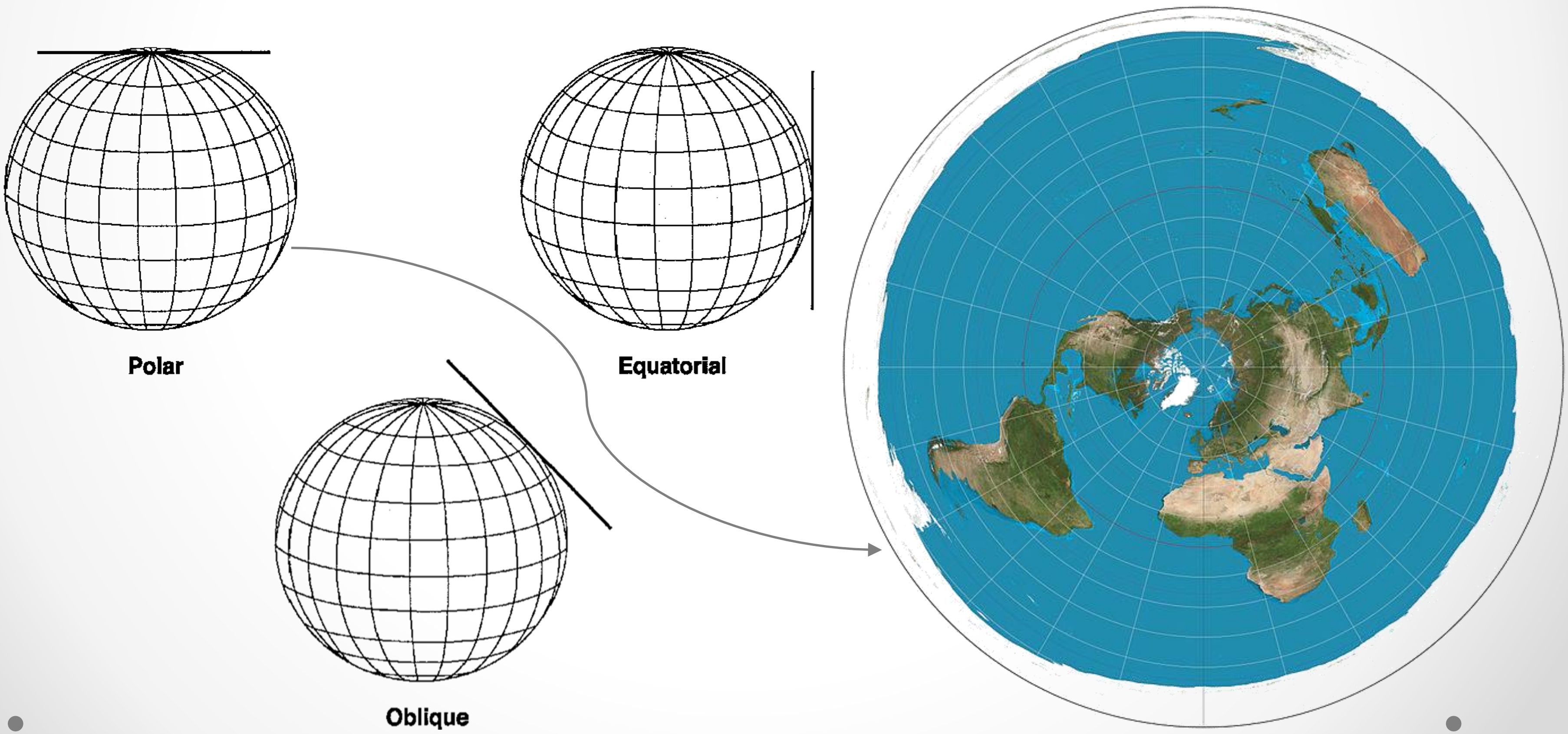


Robinson

Classification – Developable Surface

1. **Azimuthal Projection**

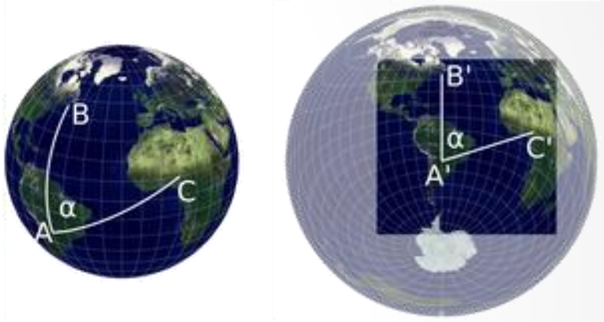
- Also called as a planar projection



Classification – Developable Surface

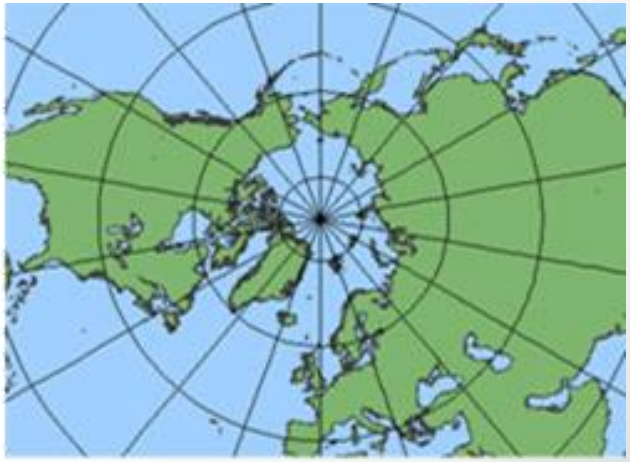
1. Azimuthal Projection

- All azimuthal projection can maintain true directions (α) from the centre point



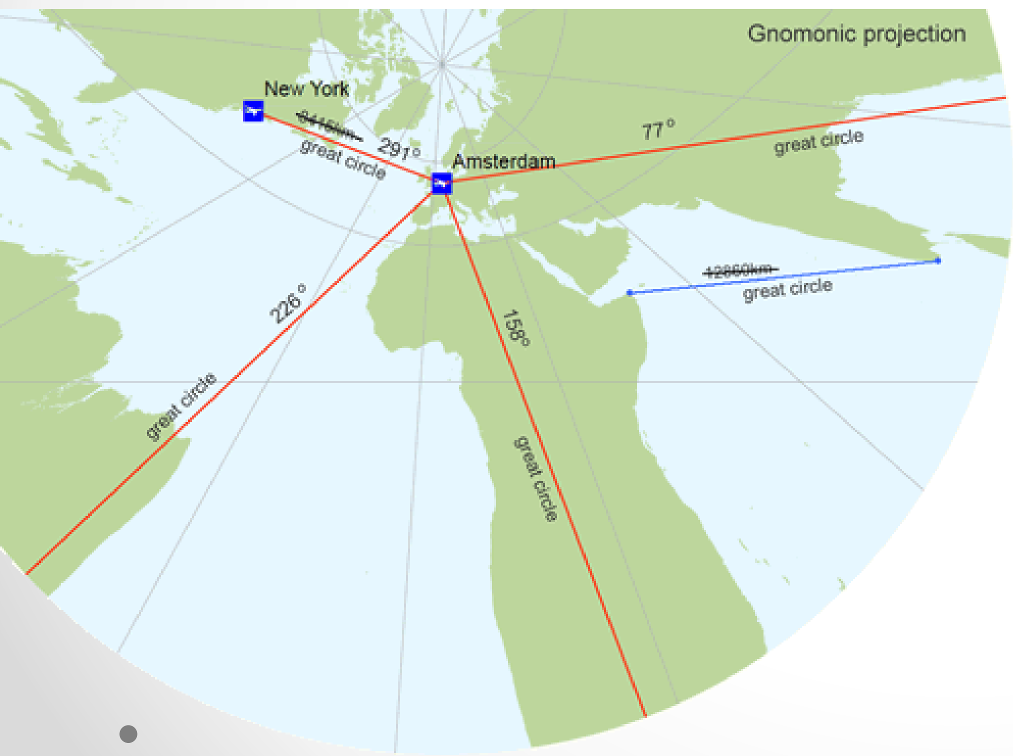
- Great circles through central point are represented as straight lines; if centre is at the poles meridians are the straight lines

- Normally one hemisphere or a portion of is represented



Gnomonic Azimuthal Projection

- Neither conformal nor equal-area
- Shortest distances between two places on a sphere - are shown as straight lines

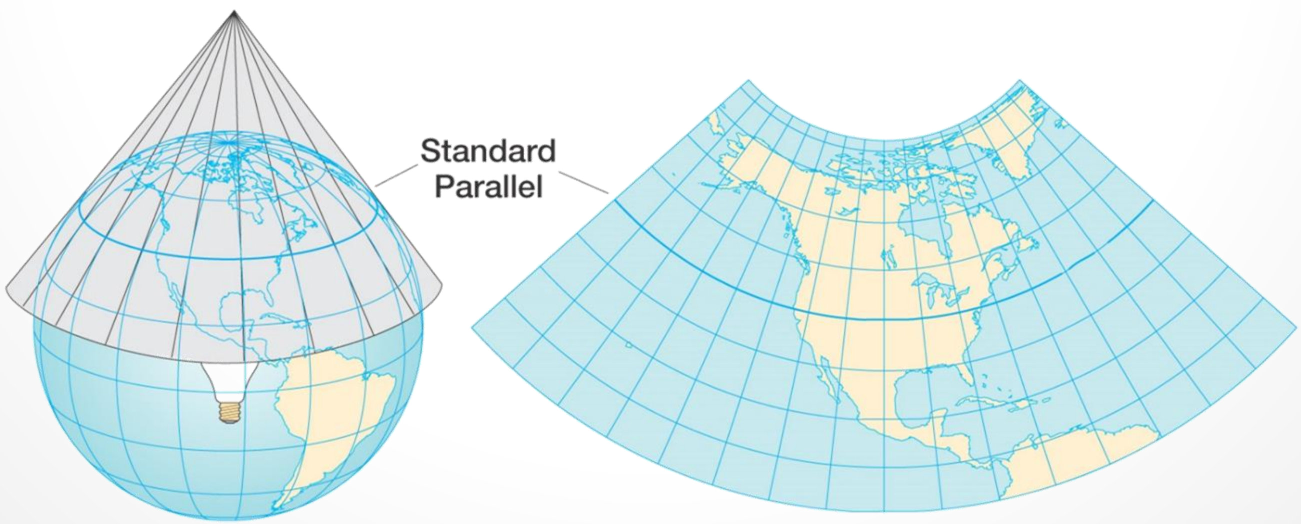


- Used widely in air-navigation
- Should orient the centre of the map at the point of interest
- Heavy distortion as we move away from the centre of the map

Classification – Developable Surface

2. Conic Projection

- Useful for representing the mid-latitudes
- The developable surface is a cone, which is "cut" along any meridian to produce the final conic projection
- Has straight converging lines for meridians and concentric circular arcs for parallels



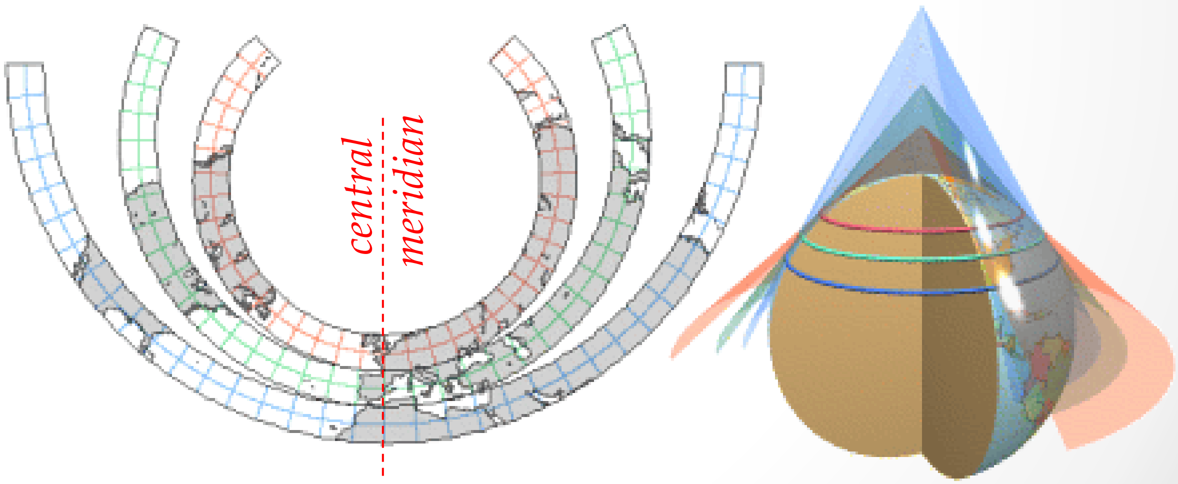
- Deviation can be minimized in a region of interest by using a secant developmental surface. Here the region of interest should lie largely between the two standard parallels

Polyconic Projection

- Many conic developable surface are used, each having a different standard parallel

- Projection unsuitable for large areas on a single sheet

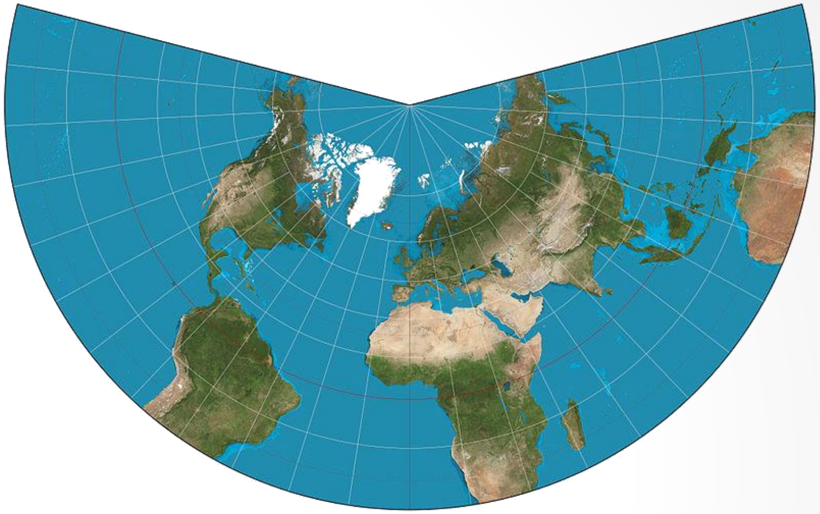
- Distortion increases as we move away from the central meridian



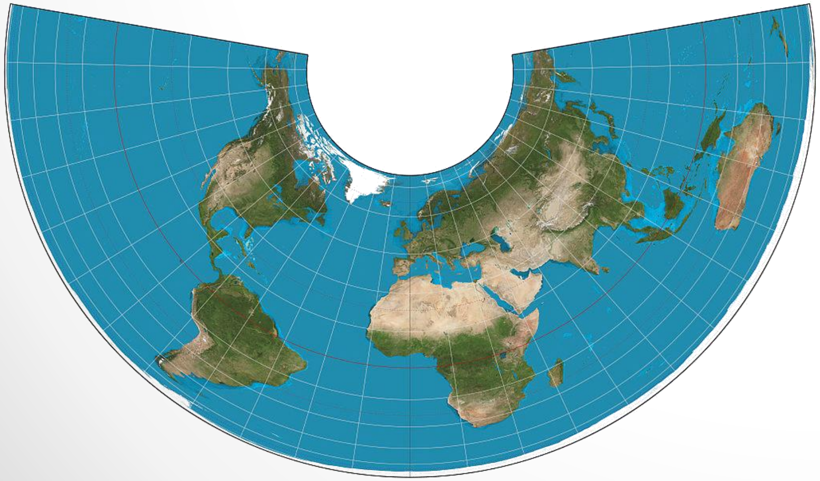
- Survey of India uses this projection system for its topographical sheets. Each sheet has a different central meridian

Lambert conformal conic projection

- Conformal
- For this, standard parallels is at 20°N and 50°N



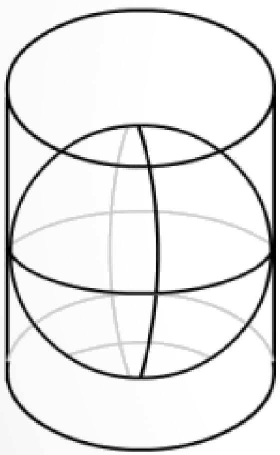
Albers equal-area projection



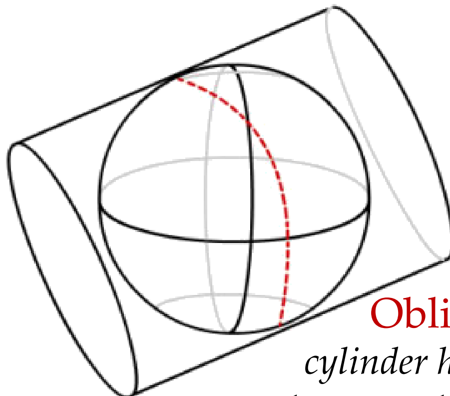
- Equal Area
- For this, standard parallels is at 20°N and 50°N

Classification – Developable Surface

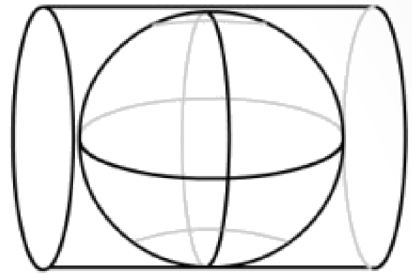
3. Cylindrical Projection



Normal
cylinder has line of tangency to the equator

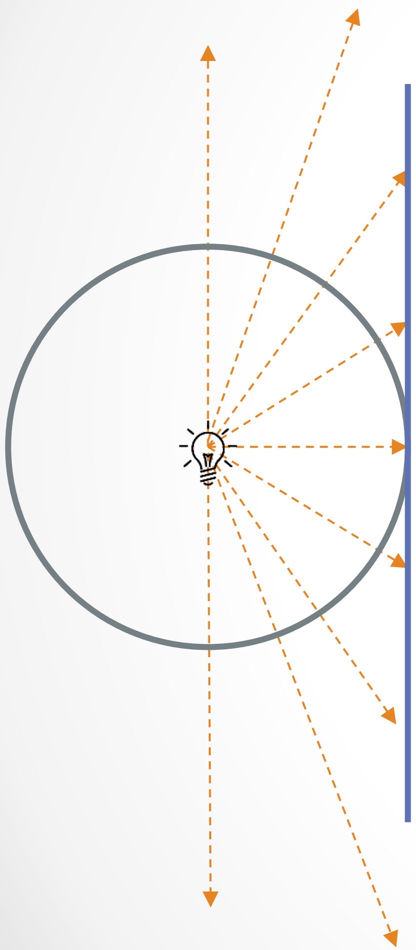


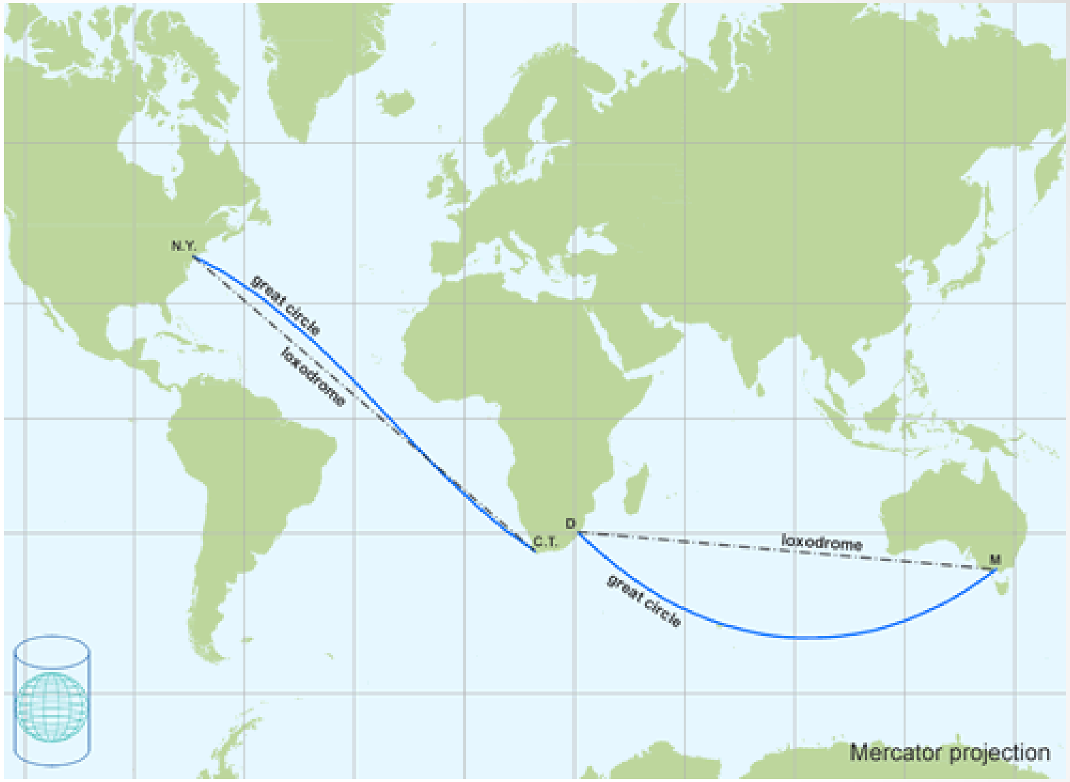
Oblique
cylinder has line of tangency to the other point on the globe



Transverse
cylinder has line of tangency to the meridian

MERCATOR PROJECTION

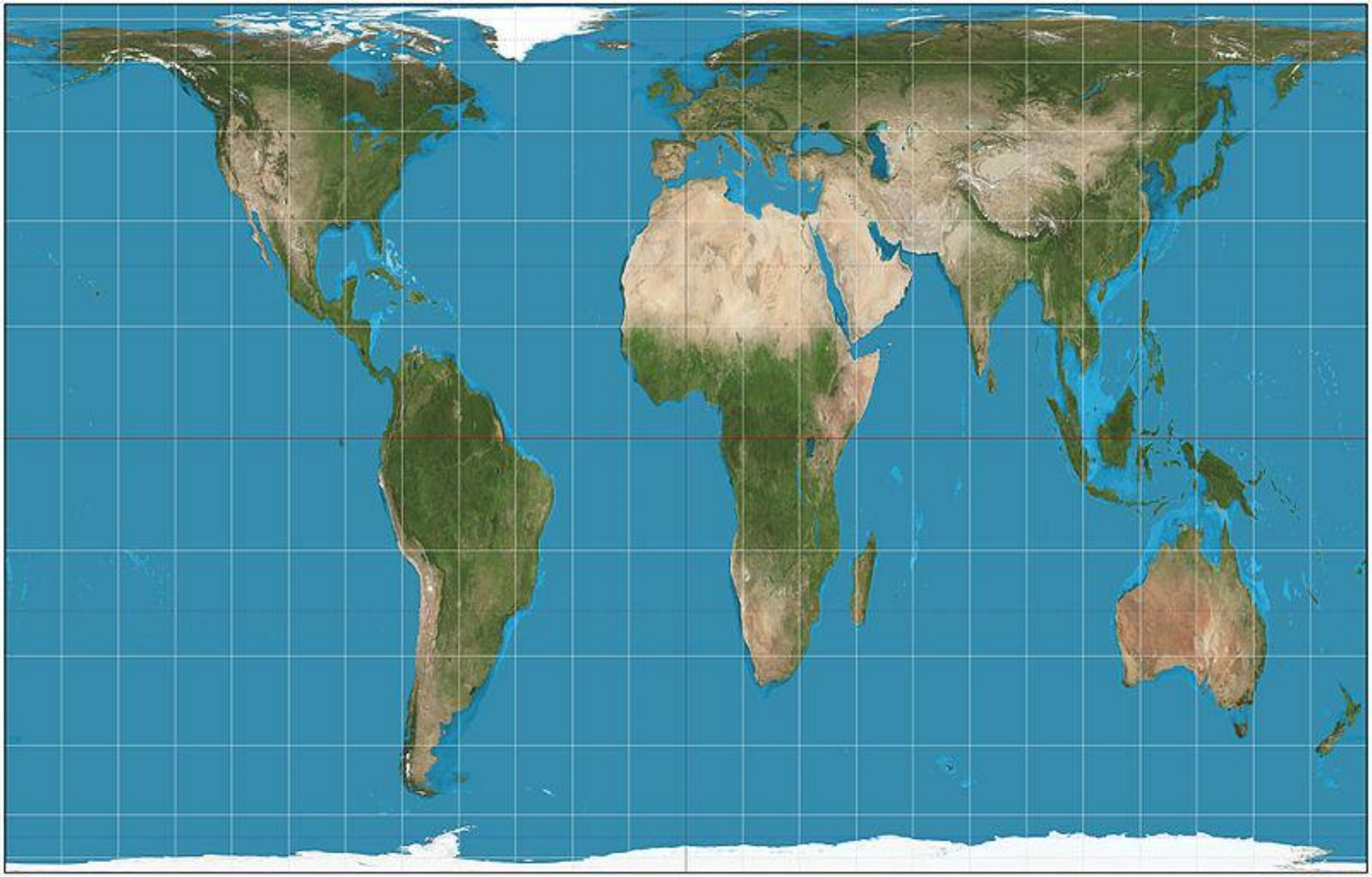




- Conformal
- Distortion is minimum near the standard parallel (here, equator)
- Distortion is very high towards the extreme ends (here, poles)

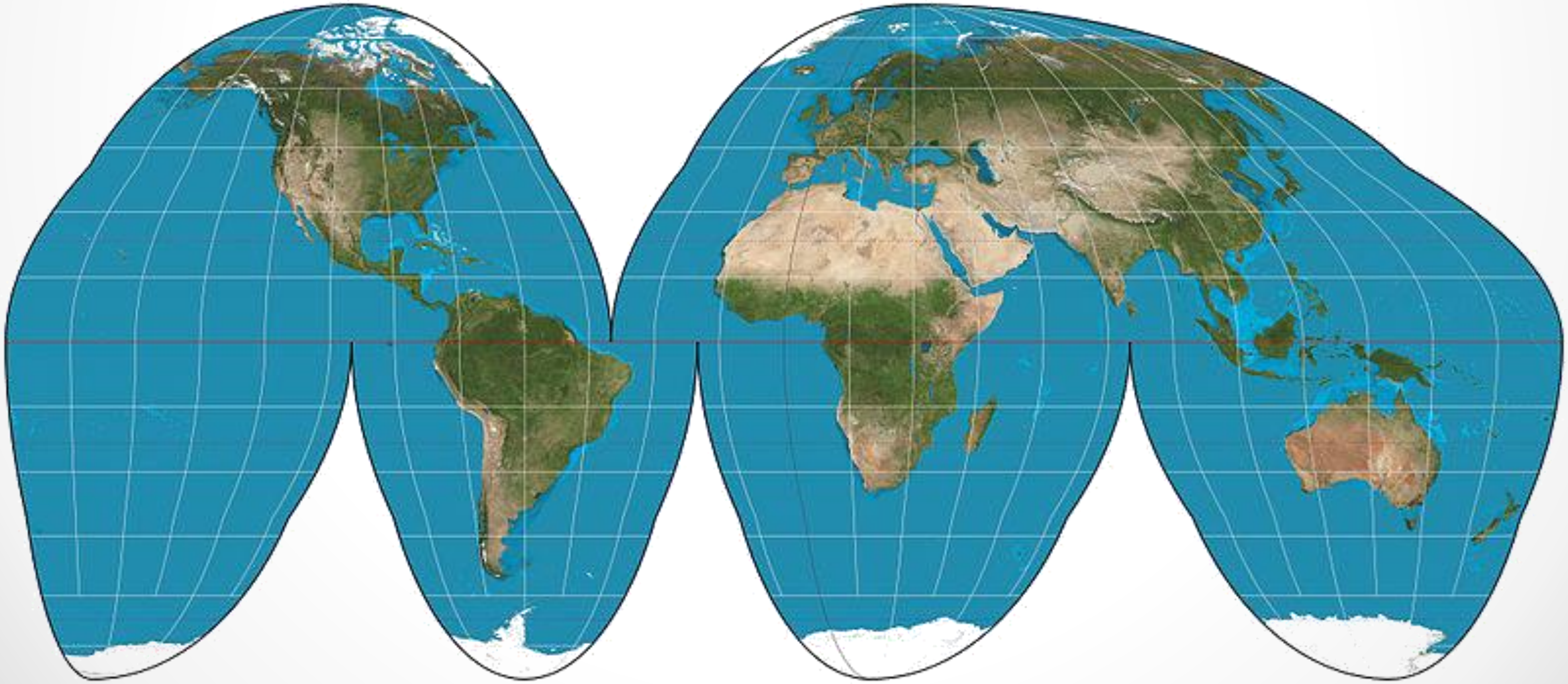
Gall–Peters equi-area projection

- Equal Area



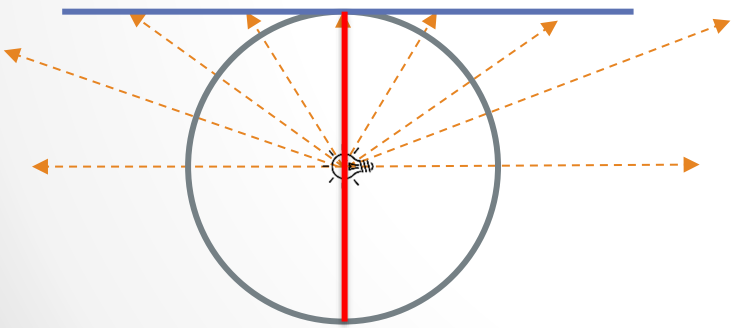
GOODE HOMOLOSINE PROJECTION

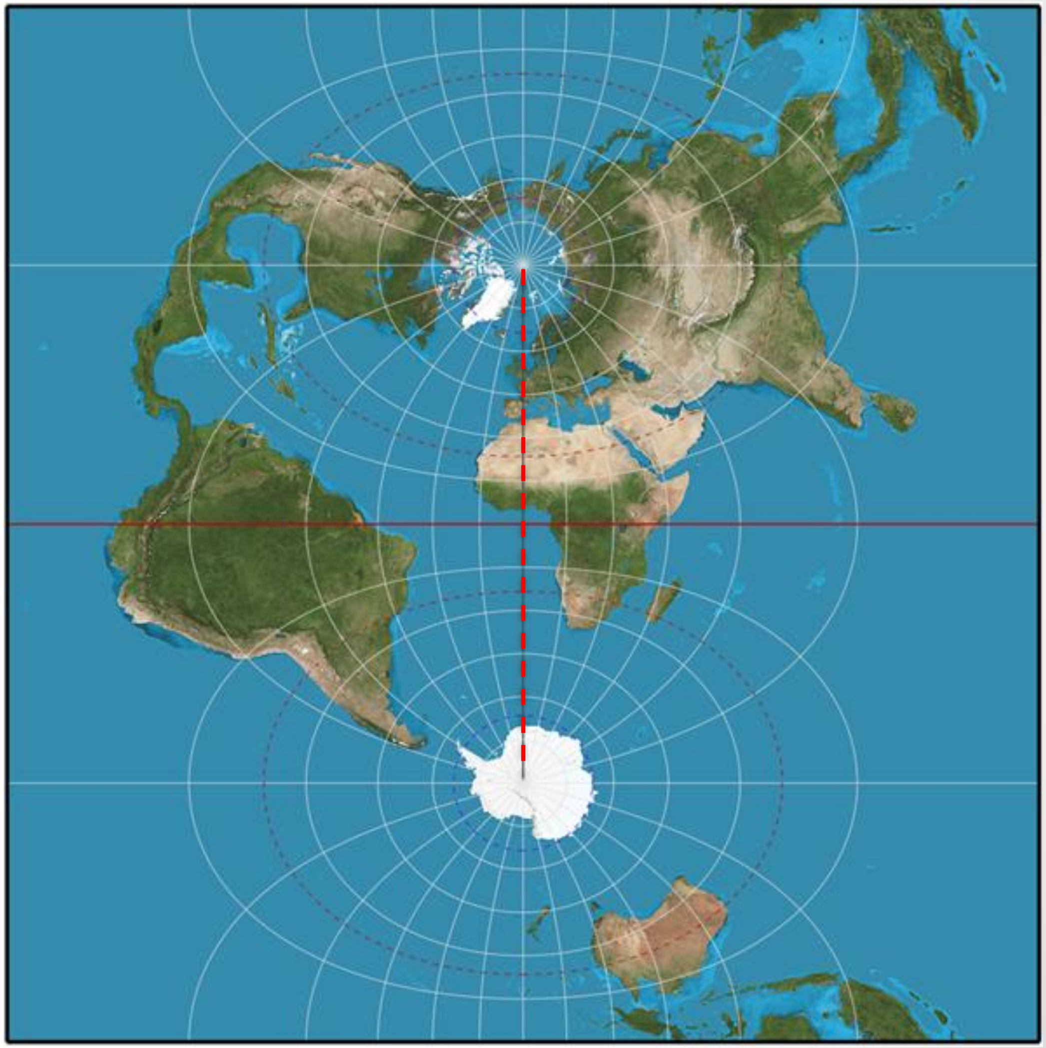
- A pseudo-cylindrical projection
- Equal Area
- Interrupted at the oceans to put an emphasis on land areas



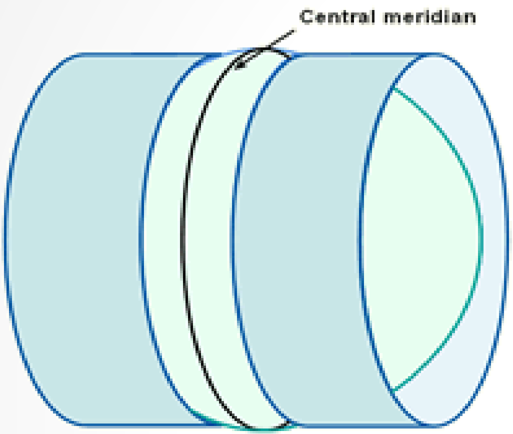
Transverse Mercator projection

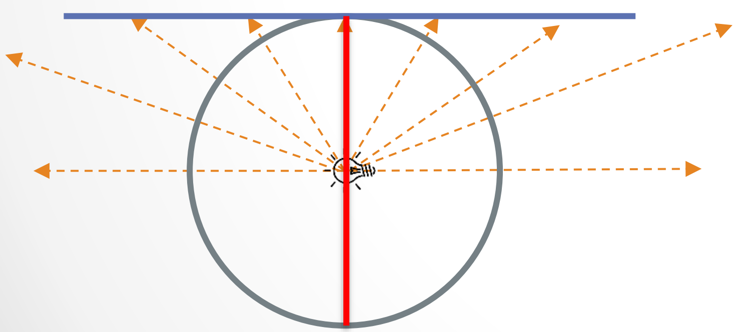
- Still Conformal
- One longitude is well represented without distortion

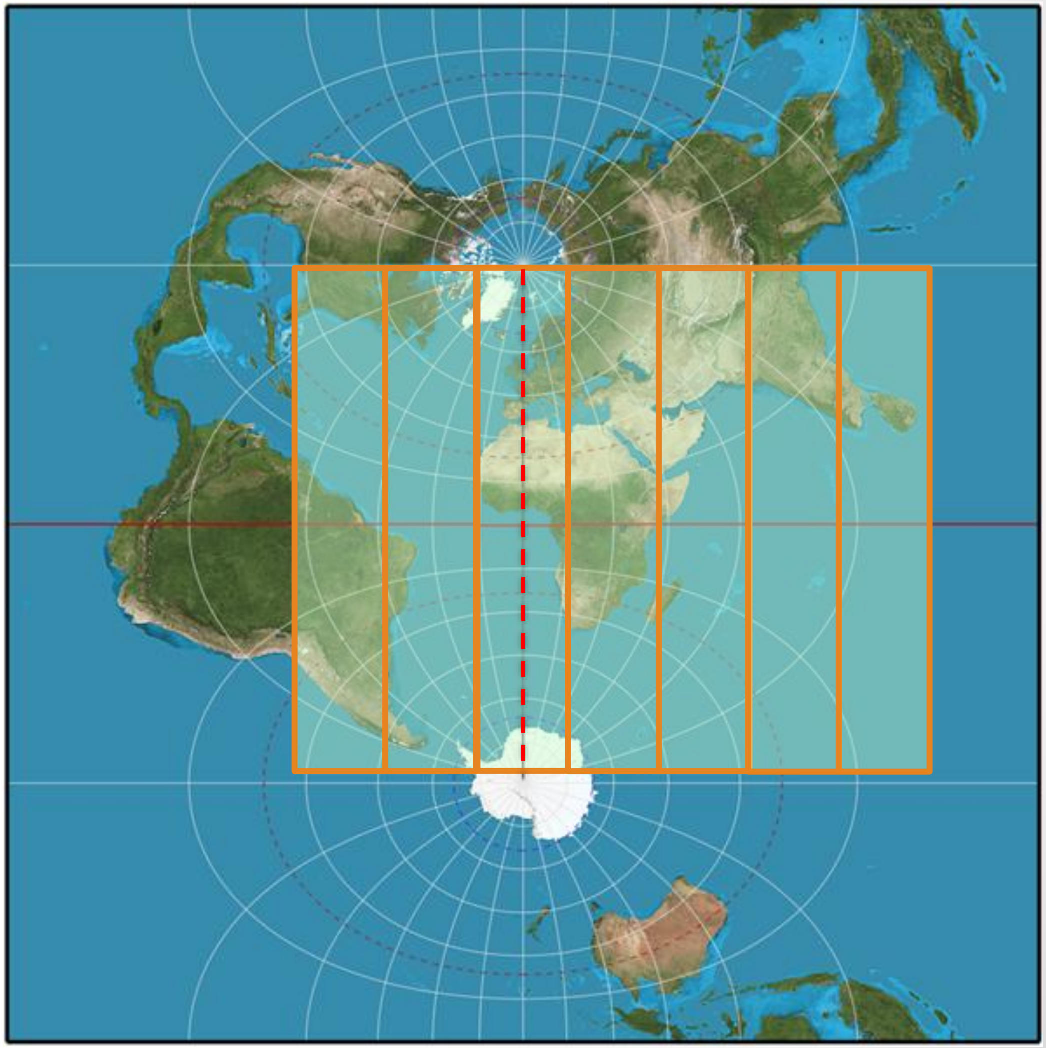




INTRODUCTION TO UTM



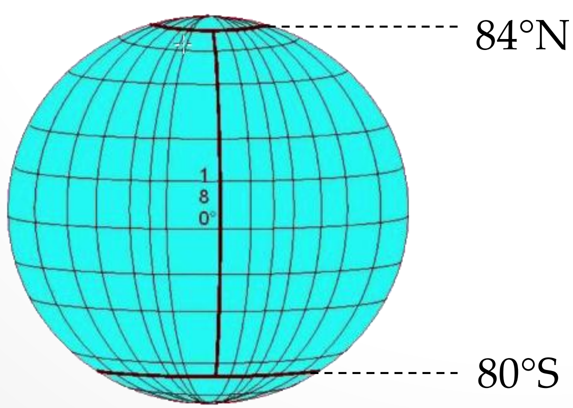


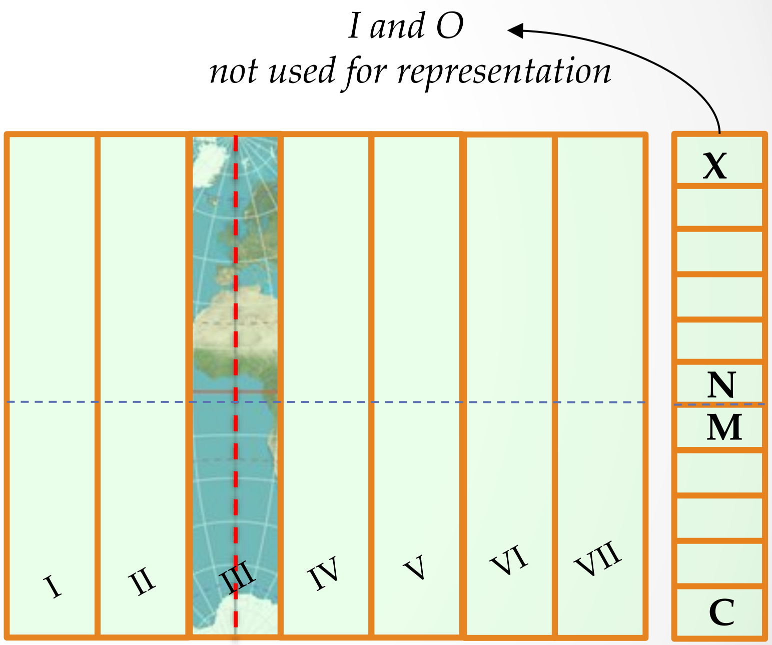


UNIVERSAL TRANSVERSE MERCATOR PROJECTION

- It divides the world into 60 narrow longitudinal zones of 6 degrees each, numbered from 1 to 60
- Each zone is a transverse cylindrical projections with a unique central meridian

Zone 1: *180° to 174° W*
Zone 60: *174° to 180° E*





- 20 latitude bands maps 80°S to 84°N, each represented by an alphabet

- Each quadrant is 6 degree wide and 8 degree tall
  <u>Exception</u>: *Quadrant 'X' which is 12 degrees tall*

- UTM uses false values (easting and northing) to derive coordinates in meters